Here is what we will cover this week:

- Why static word vectors were replaced by contextual representations, and how objectives like masked-token prediction allow models to learn powerful features without labeled data.
- Understanding scaling laws and the massive paradigm shift from bespoke, task-specific models to single, highly adaptable backbones.
- Weighing the trade-offs in accuracy and computational cost between zero-shot inference, feature extraction, full fine-tuning, and Parameter-Efficient Fine-Tuning (PEFT).
- Implementing Low-Rank Adaptation (LoRA) using the peft library to update massive models efficiently, while critically analyzing their real-world biases and failure cases.

Learning Outcomes

- Evaluate the paradigm shift toward foundation models, explaining how self-supervised learning, scaling laws, and contextual representations drive modern AI capabilities.
- Implement and compare various model adaptation strategies—ranging from zero-shot inference to parameter-efficient fine-tuning (LoRA)—to make informed engineering decisions while critically analyzing model biases and failure limitations.

Assessment